

Total No. of Questions : 8]

SEAT No. :

P-558

[Total No. of Pages : 2

[6004]-493

B.E. (Computer Engineering)
High Performance Computing
(2019 Pattern) (Semester - VIII) (410250)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.

Q1) a) Explain with diagram One-to-all broadcast on an eight-node ring with recursive doubling technique. Node 0 is the source of the broadcast. Also Explain all to one reduction with node 0 as destination. [7]

b) Explain in detail Blocking and Non-Blocking Communication Using MPI. [6]

c) Write a short note on prefix-sum operation. [4]

OR

Q2) a) What is all to all broadcast communication operation? Explain all to all broadcast on an eight node ring with step wise diagrams. (Show first two steps and last communication step). [7]

b) Explain scatter and gather communication operation with diagram. [6]

c) Explain circular shift operation? [4]

Q3) a) Explain parallel Matrix —Matrix multiplication algorithm with example? [7]

b) Explain different performance Metrics for Parallel Systems. [6]

c) Explain Minimum Execution Time and Minimum Cost Optimal Execution Time. [4]

OR

P.T.O.

- Q4)** a) What is granularity? What are effects of granularity on performance of parallel systems? [7]
b) Explain various sources of overhead in parallel systems? [6]
c) Explain “Scaling Down (downsizing)” a parallel system with example. [4]

- Q5)** a) What is CUDA? Explain different programming languages support in CUDA. Discuss any three applications of CUDA. [8]
b) Describe processing flow of CUDA-C program with diagram. [6]
c) Explain the following terms in CUDA: device, host, device code, Kernel. [4]

OR

- Q6)** a) Explain CUDA memory model. Discuss thread hierarchy. [8]
b) What is block dimension and grid dimension in CUDA? Write a CUDA kernel for addition of two vectors and explain how it will calculate addition using threads. [6]
c) What is a Kernel in CUDA? What is kernel launch? Explain arguments that can be specified in a Kernel launch. [4]

- Q7)** a) Explain odd-even transportation in bubble sort using parallel formulation. Give one stepwise example solution using odd-even transportation. [8]
b) Explain Parallel Depth First Search algorithm in detail? [6]
c) What is Kubernets? Explain its features and applications. [4]

OR

- Q8)** a) Write short notes on : [8]
i) Parallel Merge sort
ii) GPU applications
b) What are the issues in sorting on parallel computers? Explain with appropriate example? [6]
c) Explain parallel BFS algorithm in brief. [4]

Total No. of Questions: 8]

SEAT No. :

PB2256

[6263]-94

[Total No. of Pages :2

**B.E. (Computer Engineering)
HIGH PERFORMANCE COMPUTING
(2019 Pattern) (Semester-VIII) (410250)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q5 or Q6, Q7 or Q8.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Make suitable assumptions whenever necessary.

- Q1)** a) What is one-to-all broadcast? Explain it with the help of algorithm for one-to-all broadcast on hypercube. Comment on cost calculation. [7]
b) Explain scatter and gather communication operation. [6]
c) Explain circular shift operation. [4]

OR

- Q2)** a) Explain all to all broadcast and all to all reduction communication operation with example? Discuss cost analysis. [7]
b) Explain in detail Blocking and Non-Blocking Communication Using MPI. [6]
c) Comment on “Improving the Speed of Communication Operations.” [4]

- Q3)** a) Explain various sources of overhead in parallel systems. [7]
b) Show effect of granularity on performance with addition of n numbers on p processing elements. [6]
c) Explain amdahl’s and gustafson’s law. [4]

OR

- Q4)** a) Explain different performance Metrics for Parallel Systems. [7]
b) Explain parallel Matrix-Matrix multiplication algorithm with example. [6]
c) Comment on “Scalability of Parallel Systems”. [4]

P.T.O.

- Q5)** a) Draw and explain CUDA architecture in details. [8]
b) Describe processing flow of cuda along with cuda c functions. [6]
c) Write advantages and limitations of CUDA. [4]

OR

- Q6)** a) Explain how the CUDA C program executes at the kernel level with example. [8]
b) Explain cuda memory model in brief. [6]
c) Write applications of cuda. [4]

- Q7)** a) What are the issues in sorting on parallel computers, explain with example? [8]
b) Explain BFS for parallel execution & analyze its complexity. [6]
c) Write short note on Kubernets. [4]

OR

- Q8)** a) Compare an algorithm for sequential and parallel Merge sort. Analyze the complexity for the same. [8]
b) Explain Parallel Depth First Search algorithm in detail. [6]
c) Write short note on GPU Applications. [4]



Total No. of Questions : 8]

SEAT No. :

PD4589

[Total No. of Pages : 2

[6404]-94

**B.E. (Computer Engineering)
HIGH PERFORMANCE COMPUTING
(2019 Pattern) (Semester - VIII) (410250)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.

- Q1)** a) Explain Broadcast and Reduce operation with diagram. [6]
b) Explain Scatter and Gather operation. [6]
c) Write a short note on Circular shift on a mesh. [5]

OR

- Q2)** a) Explain prefix-sum operation for an eight-node hypercube. [6]
b) Explain All-to-One Broadcast and Reduction on a ring. [6]
c) Explain with example and algorithm all-to-all broadcast on 3×3 mesh. [5]

- Q3)** a) Explain performance matrices of parallel system. [6]
b) Explain Matrix – Matrix multiplication in details. [6]
c) Write a note on minimum and cost optimal execution time. [5]

OR

- Q4)** a) Write a short note on Matrix vector multiplication using: [9]
i) Row-wise 1D partitioning
ii) 2 D partitioning
iii) Comparison of 1D and 2D partitioning
b) Explain the Dense matrix algorithms. [8]
i) Matrix vector multiplication.
ii) Matrix-Matrix multiplication.

- Q5)** a) Explain CUDA architecture in details. [9]
b) Write a short note on: Managing GPU Memory. [9]

OR

P.T.O.

- Q6)** a) Modify DFS for parallel execution and analyze it's complexing. [9]
b) Explain Dijkstra's algorithm in parallel formulation. [9]

- Q7)** a) Write a short note on Parallel BFS. [4]
b) Explain the term: Communication strategies in BFS. [5]
c) Write a note on: [9]
i) Random Communication Strategy
ii) Ring Communication Strategy
iii) Blackboard Communication Strategy

OR

- Q8)** a) Explain odd-even transportation in bubble sort using suitable example. [6]
b) Write a short note on parallel formulation for CRCW PRAM. [6]
c) Explain Distributed Computing for Document Classification. [6]



Total No. of Questions : 8]

SEAT No. :

PC2381

[Total No. of Pages : 2

[6354]-498

B.E. (Computer Engineering)
HIGH PERFORMANCE COMPUTING
(2019 Pattern) (Semester - VIII) (410250)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.

- Q1)** a) Explain it with the help of algorithm for all-to-all broadcast on hypercube. [6]
b) Describe One-to-All Broadcast and All-to-One Reduction on Rings. [6]
c) Explain how non-blocking communication is finalised. [5]

OR

- Q2)** a) Explain Prefix-Sum Operations on eight-node hypercube with help of diagram. [6]
b) Explain All-to-All Broadcast and Reduction on a Ring with help of algorithm. [6]
c) What is non-blocking communication? Explain isend, irecv methods. [5]

- Q3)** a) With help of example, explain Effect of Granularity on Performance. [6]
b) Describe a situation where fine-grained parallelism might be more appropriate than coarse-grained parallelism for improving scalability [6]
c) Describe Sources of Overhead in Parallel Programs and comments on how to avoid it. [5]

OR

- Q4)** a) What are the key factors that can limit the scalability of a parallel application, and how can they be addressed? [6]
b) Explain Isoefficiency Metric of Scalability and Describe a scenario where an isoefficiency value of 1 would indicate perfect scalability. [6]
c) Enlist performance Metrics for Parallel Systems and explain any two. [5]

P.T.O.

- Q5)** a) Explain the concept of parallel processing in CUDA architecture and how it differs from traditional CPU-based computing. [8]
- b) Explain the distinction between global memory and shared memory in CUDA. How are they used differently? [6]
- c) Describe how communication managed and synchronized in CUDA. [4]

OR

- Q6)** a) Explain CUDA memory model with help of CU DA memory hierarchy.[8]
- b) Explain processing flow of CUDA along with CUDA C functions used.[6]
- c) Enlist and explain applications of CUDA. [4]

- Q7)** a) Explain parallel Bubble sort with help of algorithm. [8]
- b) Explain Kuberbets framework with help of diagram. [6]
- c) Explain how Document classification works in Distributed Computing.[4]

OR

- Q8)** a) Explain parallel Merge sort with help of algorithm. [8]
- b) Explain the concept of container orchestration in Kubernetes. How does it simplify application deployment and management? [6]
- c) Elaborate use of AI/ML in Parallel Computing. [4]



Total No. of Questions : 8]

SEAT No. :

PE-2182

[Total No. of Pages : 2

[6584]-81

B.E. (Computer Engineering)
HIGH PERFORMANCE COMPUTING
(2019 Pattern) (Semester - VIII) (410250)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.

- Q1)** a) What is one-to-all broadcast? Explain it with the help of algorithm for one-to-all broadcast on hypercube. Comment on cost calculation. [7]
- b) Explain circular shift operation? [6]
- c) Write a short note on prefix-sum operation. [4]

OR

- Q2)** a) Explain with example and algorithm all-to all broadcast on 3 * 3 mesh [7]
- b) Explain cost analysis of all to all broadcast operation. [6]
- c) Explain Scatter and Gather operations. [4]
- Q3)** a) Explain parallel Matrix—Matrix multiplication algorithm with example? [7]
- b) Explain circular shift operation on mesh and hypercube network. [6]
- c) How to improve speed of communication operation. [4]

OR

- Q4)** a) What is granularity? What are effects of granularity on performance of parallel systems? [7]
- b) Explain sources of overhead in parallel program. [6]
- c) Explain performance matrices of parallel systems. [4]

P.T.O.

- Q5)** a) What is CUDA? Explain different programming languages support in CUDA. Discuss any three applications of CUDA. [8]
- b) Describe processing flow of CUDA-C program with diagram. [6]
- c) Explain the following terms in CUDA: device, host, device code, Kernel [4]

OR

- Q6)** a) Explain compare exchange and compare-split operation on parallel Computers. [8]
- b) What are the issues in sorting on parallel computers with example? [6]
- c) Explain odd- even transportation on bubble sort using parallel formulation. [4]
- Q7)** a) Write short notes on : [8]
- i) Parallel Merge Sort ii) GPU Applications
- b) Explain communication strategies for parallel BFS. [6]
- c) What is Kuberternets? Explain its features and applications. [4]

OR

- Q8)** a) Explain the recursive decomposition in parallelizing quick sort. [8]
- b) Compare shared address and message passing formulation of quick sort. [6]
- c) Explain Parallel Depth First Search algorithm in detail? [4]

